

TECHNICAL DOCUMENTATION

LLM Agent NVIDIA

Topic discovery on financial news
with a local NVIDIA Nemotron agent

LangChain | vLLM | KMeans | LDA | Google Colab

Version 1.0

October 2026

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Dataset: magodobolo/nvidia-news (Kaggle)

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1. Summary

This project builds an **LLM agent** that explores a financial news dataset, runs the analysis through tools and explains the results. The model is NVIDIA Nemotron-3-Nano (30B-A3B, FP8), served locally on a Google Colab GPU with vLLM and orchestrated with LangChain. The agent decides when to call each tool: dataset inspection, pandas code, semantic search, KMeans clustering and LDA topic modeling with two- and three-word phrases. It then names the topics and draws the plots.

Main findings

- The agent works end to end with structured tool calling on a locally served open model.
- A large part of the feed labeled NVIDIA is templated or general market content. Before cleaning, whole LDA topics were advertising or ticker labels. After cleaning, the topics became editorial.
- KMeans does not find well-separated groups. The silhouette score stayed between 0.082 and 0.11 for 3 to 12 clusters, which points to a continuous cloud of articles.
- UMass coherence is not a reliable quality grade here. It rewards phrases that repeat together, so templated text inflates it.
- The agent can misreport results in its final summary, so the tool outputs must be read directly.

2. Business problem

Market intelligence, investor relations and equity research teams follow a constant stream of news about the companies they cover. Reading everything is slow, and keyword search only finds what the analyst already knows to look for. A large share of the feed is not editorial content: daily market wraps, advertising blocks and agency bylines repeat across hundreds of articles and hide the real themes.

The team needs a faster way to answer questions such as:

- What are the main themes in the news about NVIDIA?
- How much of the coverage is about the company itself, and how much is general market context?
- Which articles are templated reports, and which carry analysis?
- Can I ask for an analysis in plain language and get numbers I can trust?

Objectives and success criteria

- Profile the dataset and detect quality issues before modeling.
- Clean templated text so topics reflect editorial content.
- Compare two methods: KMeans on embeddings and LDA on 2- to 3-word phrases.
- Let the agent name the topics and generate the plots on request.
- A topic is acceptable when its representative headlines support its name, not only its keywords.
- The agent reports tool numbers exactly and flags topics that look like template text.

Scope. In scope: the rows with Company equal to NVIDIA in `news_v1.csv`, topic discovery, visualization and the agent workflow. Out of scope for now: sentiment, topic evolution over time and any investment recommendation. The analysis describes what the news says. It does not predict prices.

3. Data

Source: *magodobolo/nvidia-news* on Kaggle. The dataset has two files. Only the first is used so far.

File	Columns	Status
news_v1.csv	Company, Headline, Summary, Date	Main analysis data
nvidia_sentimental_analysis.csv	Date, Summary, sentiment_rf, sentiment_nn	Available, not yet explored

Profile of news_v1.csv

Item	Value
Rows	1047
Null values	0 in every column
Companies	NVIDIA (911) and SPOTIFY (136)
Period	2015-08-06 to 2024-06-14
Language	Mostly Portuguese
Date column	Stored as text, needs pd.to_datetime for time analysis

Data quality observations

- **The company label looks like a feed label.** Of the 911 rows labeled NVIDIA, 28 never use the word nvidia in the headline or summary, and only 10 of those 28 cite NVDA, Huang or Jensen. Their headlines suggest general market bulletins and articles about other chipmakers.
- **Agency bylines.** 68 summaries end with a credit line such as a reporting credit in parentheses, which put journalist names into the keywords.
- **Repeated sentences.** 43 sentences appear in at least 8 articles. They are mainly InvestingPro advertising and fixed blocks of the market wrap reports.
- **Daily wraps.** 71 rows matched the headline pattern of daily opening and closing reports.
- **Promotional text.** 133 summaries contain the words cupom, investingpro or protips.

4. Architecture

Data flows from the Kaggle dataset to cleaned texts, then to embeddings and phrase counts. The agent talks to the model through the OpenAI-compatible API exposed by vLLM and calls tools that work on the DataFrame and on those representations.

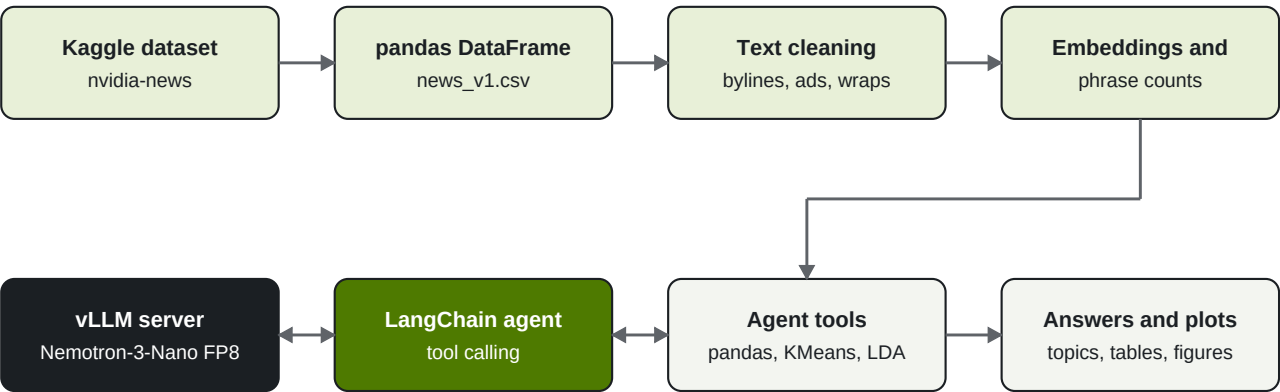


Figure 1. Data and agent architecture.

Component	Detail
Environment	Google Colab, G4 GPU (NVIDIA RTX PRO 6000 Blackwell Server Edition, about 97 GB of VRAM)
LLM	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8, served by vLLM
Orchestration	LangChain 1.x with create_agent, ChatOpenAI pointing to the local server
Semantic search	sentence-transformers/all-MiniLM-L6-v2 with an in-memory vector store
Topic embeddings	paraphrase-multilingual-MiniLM-L12-v2, normalized
Topic models	scikit-learn KMeans and LatentDirichletAllocation
Visualization	matplotlib, t-SNE projection to 2D

5. Environment and setup

Verified versions

Package	Version	Package	Version
vllm	0.30.0	langchain	1.4.3
torch	2.13.0 (CUDA 13.0)	langchain-core	1.6.6
torchvision	0.28.0	langchain-openai	1.6.7
transformers	5.13.1	langgraph	1.2.12
Python	3.12	NVIDIA driver	580.82.07

Prerequisites

- A Colab account with access to a high-memory GPU. The tests used the G4.
- Colab secrets named HF_TOKEN and kaggle_api. A nvidia_api secret exists but is not used in this flow.
- Acceptance of the model terms on Hugging Face, when requested.
- About 30 GB of free disk for the first model download.

Run order

- Install the dependencies in one command and **restart the session**. Without the restart, the old langchain-core stays loaded in memory and the langchain-openai import fails.
- Run the cells for secrets, dataset, text cleaning and embeddings.
- Start the vLLM server in the background and wait for the ready message.
- Create the tools and the agent, then ask the questions.

```
vllm serve nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8 \
  --served-model-name model --port 8000 \
  --max-model-len 32768 --gpu-memory-utilization 0.70 \
  --trust-remote-code --enable-auto-tool-choice \
  --tool-call-parser qwen3_coder \
  --reasoning-parser-plugin nano_v3_reasoning_parser.py \
  --reasoning-parser nano_v3
```

The reasoning parser plugin is downloaded from the model repository. The first start downloads the weights and may compile kernels, so it can take several minutes. Memory utilization was 0.80 in the first runs and 0.70 later, to leave room for other models on the same GPU.

6. Agent design

Tool	What it does
dataset_overview	Shows shape, columns, dtypes, null counts and sample rows of the DataFrame
run_pandas	Runs pandas code on a copy of df and returns the last expression
nvidia_news_search	Semantic search over the news texts
cluster_topics	KMeans clusters with size, keywords, silhouette and representative headlines
plot_clusters	2D scatterplot (t-SNE) of the KMeans clusters
lda_topics	LDA with 2- to 3-word phrases, coherence and representative headlines
plot_lda_topics	Bar chart with the top phrases of each topic
plot_lda_scatter	2D scatterplot of the news colored by dominant topic

System prompt principles

- Inspect the dataset first and never invent numbers. Counts and statistics come from run_pandas.
- Name each topic from its representative headlines first and its phrases second.
- Use the same topic names in the plot calls and in the final answer.
- Say so when the phrases of a topic look like advertising, ticker labels, weekdays or template text.
- Copy coherence values and counts exactly from the tool result, and never present coherence as a quality ranking.

Observed behavior

In the first exploratory run the agent called dataset_overview, then run_pandas several times, then the semantic search, and answered with a coherent profile (1047 rows, 911 NVIDIA and 136 Spotify articles, no nulls, 2015 to 2024). Four of nine run_pandas calls returned no value, because the model wrote an expression without assigning it to the result variable. The fix was to evaluate the last expression of the submitted code automatically.

Security note. run_pandas executes model-generated code with exec inside the Colab VM. That is acceptable for personal use. Do not expose this agent to third parties without a real sandbox, because the process can read environment variables such as the Hugging Face token.

7. Preprocessing

Text cleaning steps applied before topic modeling:

- Removal of credit lines in parentheses, which polluted keywords with journalist names.
- Removal of the 43 repeated sentences, found by masking numbers and counting sentences that appear in at least 8 articles.
- Removal of links, numbers and exchange labels such as a ticker in parentheses.
- Headline deduplication after stripping numbers, so repeated daily reports count once.
- Exclusion of rows that match the daily wrap headline pattern from the LDA input.
- Stop words in Portuguese and English, plus weekdays, company names and promotional terms.

Stage	NVIDIA rows in LDA input
Labeled NVIDIA	911

Stage	NVIDIA rows in LDA input
After headline deduplication	762
After excluding daily wraps	741

71 rows matched the daily wrap pattern across the whole dataset. Many of them were already removed by deduplication, so 21 additional NVIDIA rows left the input. The 133 rows with promotional words stay in the input, and only the promotional words were added to the stop list.

8. Methods

KMeans on embeddings

- Text per article: headline plus the first 500 characters of the summary.
- Normalized multilingual sentence embeddings, so KMeans behaves like cosine clustering.
- KMeans with `n_init` 10 and `random_state` 42.
- Keywords per cluster: TF-IDF over the concatenated texts of each cluster, with Portuguese and English stop words.
- Representative headlines: the three articles closest to each cluster center.
- Quality check: silhouette score. The 2D map uses t-SNE.

LDA on phrases

- Phrases of 2 to 3 words (ngram range 2 to 3), at least 3 letters per token, `min_df` 5 and `max_df` 0.5.
- Low priors (`doc_topic_prior` 0.1, `topic_word_prior` 0.01) favor few topics per text and few phrases per topic.
- Batch learning, 40 iterations, `random_state` 42.
- Per-topic UMass coherence on the top 10 phrases. Closer to zero means the phrases co-occur more.
- Share of texts whose dominant topic weight is above 0.5.
- The scatterplot applies t-SNE to the square root of each document's topic distribution, which approximates the Hellinger distance. Point size grows with the weight of the dominant topic.

Why two methods. KMeans groups whole articles by meaning. LDA describes each article as a mix of topics defined by phrases, so its topics are easier to read. They can disagree, and the comparison shows how much each result depends on the method.

9. Results

9.1 LDA topics (8 topics, phrases of 2 to 3 words)

Input: 741 articles and 3000 phrases. Mean UMass coherence: **-1.71**. Share of texts with a dominant topic above 0.5: **0.80**. The readings are interpretations of the phrases and of the representative headlines. Phrases are shown as extracted, in Portuguese.

Topic	Texts	Reading	Highlighted phrases	Example headline
0	52	Wall Street indexes, chips and rates	wall street, pontos enquanto, taxa juros	S&P 500 e Nasdaq sobem com salto de ações de chips e megacaps
1	89	Analyst recommendations and price targets	lucro ação, itaú bba, goldman sachs	Pode subir mais: Barclays eleva preço-alvo e recomenda compra em Nvidia
2	57	Market agenda and interest rates	contratos futuros, presidente fed, horário Brasília	Ações na Europa registram ganhos leves em meio à incerteza sobre juros

Topic	Texts	Reading	Highlighted phrases	Example headline
3	53	Stock rally, AI and China restrictions	restrições exportação, chips china, titãs tecnologia	Conheça a ação que subiu mais que Nvidia este mês
4	212	Earnings, price targets and data centers	preço alvo, data centers, morgan stanley	4 destaques dos analistas: Nvidia dispara após passar por cima das estimativas
5	108	NYSE and Nasdaq closes, Fed and monetary policy	dow jones, federal reserve, política monetária	Bolsas de NY fecham em queda após inflação acima do esperado
6	130	NVIDIA market value	bilhões dólares, valor mercado, fabricante chips	Nvidia ultrapassa Apple e se torna a segunda empresa mais valiosa do mundo
7	40	Market closes and export restrictions (mixed)	subiu pontos, fecharam alta, primeiro trimestre	EUA restringem exportações de chips de IA da Nvidia e AMD para Oriente Médio

Topics 1, 4 and 7 deserve care. Topic 1 still carries residual promotional phrases, topic 4 is broad (212 texts) and topic 7 mixes two subjects.

Effect of cleaning on the metrics

Run	Rows	Topics	UMass coherence	Dominant topic share
Raw text (ads and bylines present)	762	6	-1.31	0.88
After removing repeated sentences	762	6	-1.56	0.90
After also removing daily wraps	741	6	-1.83	0.85
After also removing daily wraps	741	8	-1.71	0.80

Coherence got worse at each cleaning step, while the topics got more editorial. In the raw run, one topic was made of advertising phrases and had the best coherence of all. Repeated template text co-occurs in every copy, which makes coherence look good. This is the main reason not to read UMass as a quality grade in this dataset.

Number of topics, run on 762 rows before removing daily wraps

Topics	4	6	8	10	12
UMass coherence	-1.84	-1.56	-1.49	-1.70	-1.68
Dominant topic share	0.95	0.90	0.84	0.82	0.76

The dominant share falls as the number of topics grows, which is expected. The coherence gap between 6 and 8 topics is small for about 700 short texts and a single seed, so the choice should come from reading the headlines of each topic.

9.2 KMeans clusters (6 clusters, 911 NVIDIA rows)

This run used embeddings generated before the agency bylines were removed. The names were assigned by the agent from keywords and sample headlines.

Cluster	Texts	Agent name (translated)	Topical keywords	Example headline
0	212	Price targets and recommendations	alvo, compra, recomendação, consenso, ubs	Nvidia: ainda vale a pena investir com rali pós-balanço?
1	125	Daily market moves (templated)	encerramento, avançou, recuou, inalteradas	E.U.A. - Ações fecharam o pregão inalteradas e o Índice Dow Jones avançou 0,26%
2	166	Funds and large tech	fundos, compra, softbank, magnificas	Além de Nvidia: estas são as ações de tecnologia mais promissoras para 2024
3	57	Megacaps and market	megacaps, negociava, recuava, trump	Wall St abre em baixa antes de resultados da Nvidia e ata do Fed

Cluster	Texts	Agent name (translated)	Topical keywords	Example headline
4	157	New processors and AI	processamento, huang, processadores, jogos	Nvidia lançará chip de IA voltado para a China no 2º tri, dizem fontes
5	194	Interest rates, inflation and tariffs	powell, trump, tarifas, juros	Futuros de Wall Street caem em meio à incerteza de corte de juros do Fed

Journalist names that also appeared in the raw keyword output were left out of this table. Clusters 1 and 3 describe almost the same thing, a Wall Street market summary, so 6 clusters may be too many.

Clusters	3	4	5	6	7	8	9	10	11	12
Silhouette	0.110	0.097	0.097	0.102	0.094	0.097	0.101	0.105	0.083	0.082

The silhouette is flat and low from 3 to 12 clusters. The articles form a continuous cloud with no well-separated groups, so the number of clusters should be chosen by interpretability.

9.3 Plots

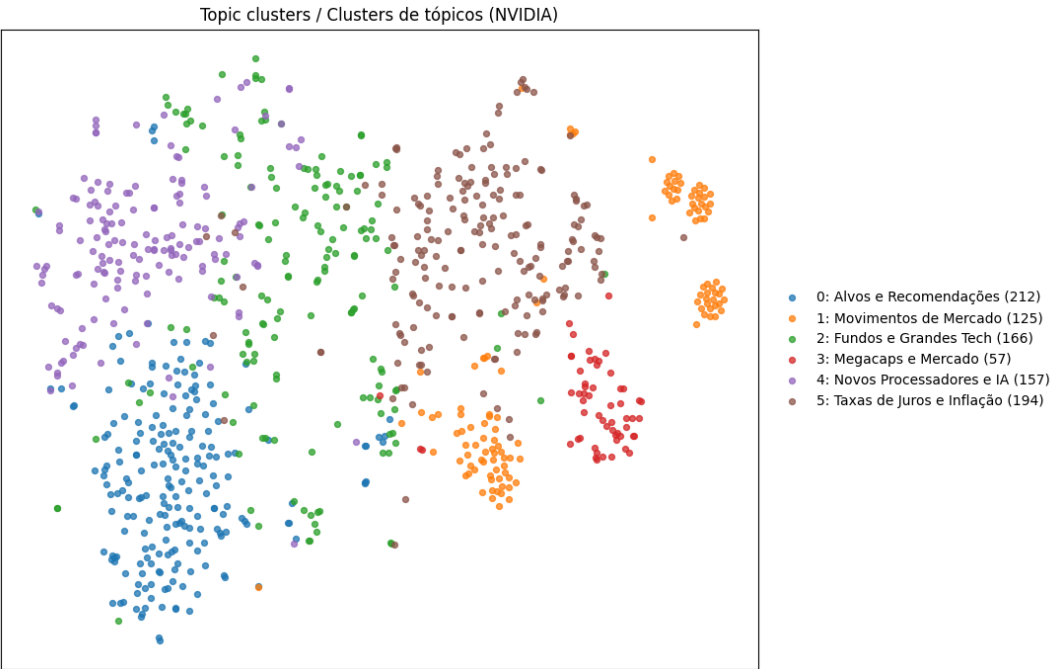


Figure 2. KMeans clusters on a 2D t-SNE map (911 NVIDIA articles). Names assigned by the agent.

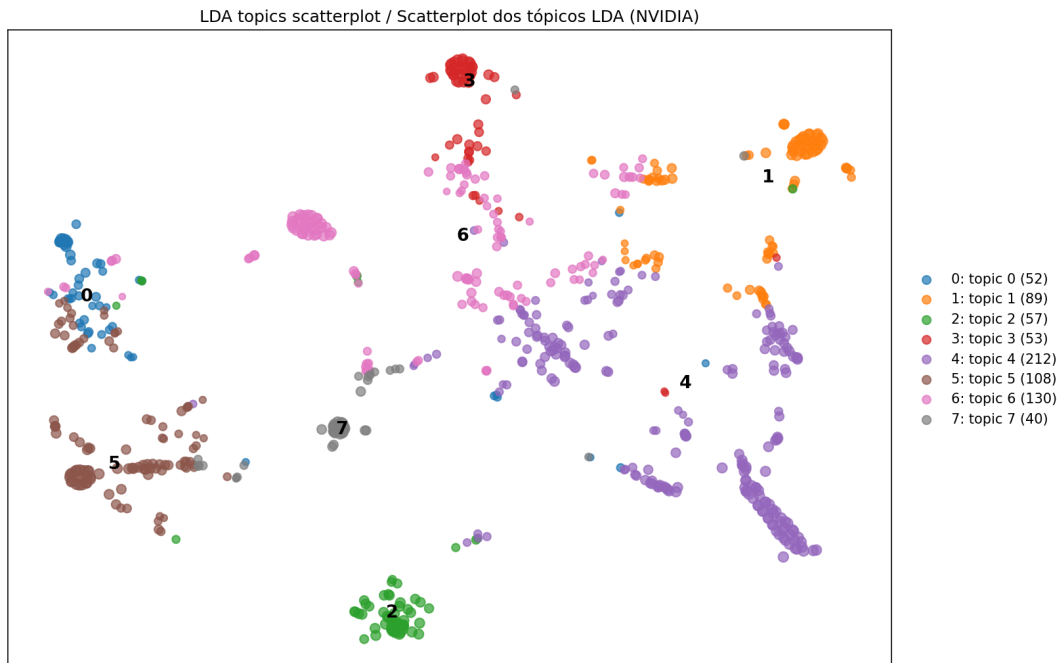


Figure 3. LDA topics on a 2D t-SNE map (741 articles, 8 topics). The number marks the median of the points of each topic and can fall in empty space when a topic is fragmented.

How to read the plots

- In t-SNE, local neighborhoods are meaningful. The distance between far-apart groups is not.
- Dense blocks in the LDA plot can be near-identical texts that land on the same point.
- Points between two blocks are articles whose topic mix is split, which helps to find poorly separated topics.
- The LDA prior is low, so most texts end up with one dominant topic. Clean blocks do not prove better topics.

10. Issues found and fixes

Issue	Cause	Fix
ImportError: ModelAPIError when importing langchain_openai	The old langchain-core stayed loaded in memory after the upgrade	Restart the Colab session after installing packages
PyTorch and TorchAudio built with different CUDA versions	vLLM upgraded torch, and the published torchaudio does not follow that version	Uninstall torchaudio, which the project does not use
Connection refused on port 8000	The vLLM server was not running	Read the server log, then start the server again
No supported CUDA architectures found for major versions [12]	JIT compilation of FlashInfer FP8 kernels on the Blackwell GPU	Set FLASHINFER_CUDA_ARCH_LIST for the server process, or try the BF16 weights of the model
Plots not shown by the agent	The agent did not call the plot tool, or inline display failed	Call the tools directly and display the saved PNG files
Empty results from run_pandas	The model did not assign the final value to the result variable	Evaluate the last expression of the submitted code automatically

11. Limitations

- **Topic stability.** LDA on about 700 short texts is unstable. Change the random seed and check whether the main topics repeat.
- **Overlapping clusters.** The low silhouette means KMeans clusters overlap.
- **Residual templated text.** Variations of advertising still appear in one topic, and two topics are broad or mixed.
- **Company label.** The NVIDIA label indicates the feed, and part of the news covers the market in general.
- **Agent summaries.** The final answer can contain mistakes, for example ranking topics by coherence or describing a bar chart as a 2D map. The tool outputs are the source of truth.
- **Interpretations.** Topic names are readings of phrases and headlines, not ground truth.
- **Not yet validated.** The KMeans run still uses embeddings built before the byline cleanup.

12. Next steps

- Classify sentiment with prompt engineering (zero-shot, few-shot and rubric) and compare with `sentiment_rf` and `sentiment_nn`.
- Track the evolution of each topic over the years after converting the Date column.
- Flag articles that name NVIDIA in the headline and compare topics with and without that filter.
- Test HDBSCAN or BERTopic, which allow articles with no assigned topic.
- Rerun KMeans with embeddings built from the cleaned text.
- Label about 50 articles by hand to measure agreement instead of relying on proxy metrics.

13. References and credits

- Dataset: [magodobolo/nvidia-news](#), Kaggle.
- Model: [nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-FP8](#), Hugging Face.
- Libraries: `vLLM`, `LangChain`, `LangGraph`, `scikit-learn`, `sentence-transformers`, `pandas`, `matplotlib`.
- Cover image of the repository README: abstract low poly banner from [Magnific](#). Check the license and the attribution terms before publishing.